

Hi This is my datat analytics project on chocolate sales data set, that I found in kaggle.

This project analyzes sales performance for a chocolate company using Python, Pandas, and data visualization techniques. The goal is to uncover meaningful business insights by exploring revenue patterns, product performance, shipping behavior, and customer demand across countries and time periods.

```
In [86]: import kagglehub
path = kagglehub.dataset_download("saidaminsaidaxmadov/chocolate-sales")
print("Path to dataset files:", path)
```

Path to dataset files: C:\Users\X\.cache\kagglehub\datasets\saidaminsaidaxmadov\chocolate-sales\versions\2

```
In [7]: import os
os.listdir(r"C:\Users\X\.cache\kagglehub\datasets\saidaminsaidaxmadov\chocolate-
```

```
Out[7]: ['Chocolate Sales (2).csv',
'ChocolateSales.pbitt',
'ChocolateSales.pbix',
'ChocolateSales.pdf']
```

```
In [75]: import pandas as pd
df = pd.read_csv(
    r"C:\Users\X\.cache\kagglehub\datasets\saidaminsaidaxmadov\chocolate-sales\
df.head()
```

```
Out[75]:
```

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
0	Jehu Rudeforth	UK	Mint Chip Choco	04/01/2022	\$5,320.00	180
1	Van Tuxwell	India	85% Dark Bars	01/08/2022	\$7,896.00	94
2	Gigi Bohling	India	Peanut Butter Cubes	07/07/2022	\$4,501.00	91
3	Jan Morforth	Australia	Peanut Butter Cubes	27/04/2022	\$12,726.00	342
4	Jehu Rudeforth	UK	Peanut Butter Cubes	24/02/2022	\$13,685.00	184

```
In [11]: df.shape
df.info()
df.columns
df.describe
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3282 entries, 0 to 3281
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Sales Person          3282 non-null  object
1   Country               3282 non-null  object
2   Product               3282 non-null  object
3   Date                  3282 non-null  object
4   Amount                3282 non-null  object
5   Boxes Shipped         3282 non-null  int64
dtypes: int64(1), object(5)
memory usage: 154.0+ KB
```

```
Out[11]: <bound method NDFrame.describe of
Product      Date \
0      Jehu Rudeforth      UK      Mint Chip Choco  04/01/2022
1      Van Tuxwell        India    85% Dark Bars  01/08/2022
2      Gigi Bohling       India    Peanut Butter Cubes  07/07/2022
3      Jan Morforth      Australia Peanut Butter Cubes  27/04/2022
4      Jehu Rudeforth      UK      Peanut Butter Cubes  24/02/2022
...      ...      ...      ...      ...
3277   Karlen McCaffrey   Australia Spicy Special Slims  17/05/2024
3278   Jehu Rudeforth     USA      White Choc  07/06/2024
3279   Ches Bonnell      Canada    Organic Choco Syrup  26/07/2024
3280   Dotty Strutley     India      Eclairs  28/07/2024
3281   Karlen McCaffrey   India    70% Dark Bites  23/05/2024

      Amount  Boxes Shipped
0      $5,320.00      180
1      $7,896.00      94
2      $4,501.00      91
3     $12,726.00     342
4     $13,685.00     184
...      ...      ...
3277   $5,303.58     354
3278   $7,339.32     121
3279    $616.09     238
3280   $2,504.62     397
3281   $5,915.87     355
```

```
[3282 rows x 6 columns]>
```

```
In [13]: #Cleaning the Data
#From df.describe : 3 Date 3282 non-null object
df['Date']=pd.to_datetime(df['Date'],dayfirst=True, errors='coerce')
df['Date'].dtype
df=df.sort_values('Date')
df.head()
```

Out[13]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
1012	Dotty Strutley	USA	Raspberry Choco	2022-01-03	\$3,437.00	46
253	Marney O'Brien	Australia	Smooth Sliky Salty	2022-01-03	\$3,745.00	170
89	Andria Kimpton	Canada	Caramel Stuffed Bars	2022-01-03	\$7,154.00	348
187	Dotty Strutley	UK	99% Dark & Pure	2022-01-03	\$9,310.00	282
185	Camilla Castle	UK	White Choc	2022-01-03	\$9,982.00	187

```
In [15]: df.isna().sum()
df['Amount'] = df['Amount'].str.replace('$', '', regex=False)
df['Amount'] = df['Amount'].str.replace(',', '', regex=False)
df['Amount'].str.strip()
df['Amount'] = pd.to_numeric(df['Amount'], errors='coerce')
```

```
In [17]: df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['MonthName'] = df['Date'].dt.strftime('%B')
df['Quarter'] = df['Date'].dt.quarter

df.head()
```

Out[17]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped	Year	Month	MonthName
1012	Dotty Strutley	USA	Raspberry Choco	2022-01-03	3437.0	46	2022	1	January
253	Marney O'Brien	Australia	Smooth Sliky Salty	2022-01-03	3745.0	170	2022	1	January
89	Andria Kimpton	Canada	Caramel Stuffed Bars	2022-01-03	7154.0	348	2022	1	January
187	Dotty Strutley	UK	99% Dark & Pure	2022-01-03	9310.0	282	2022	1	January
185	Camilla Castle	UK	White Choc	2022-01-03	9982.0	187	2022	1	January



```
In [19]: df['Amount'].dtype
df[df['Amount'].isna()]
```

Out[19]:

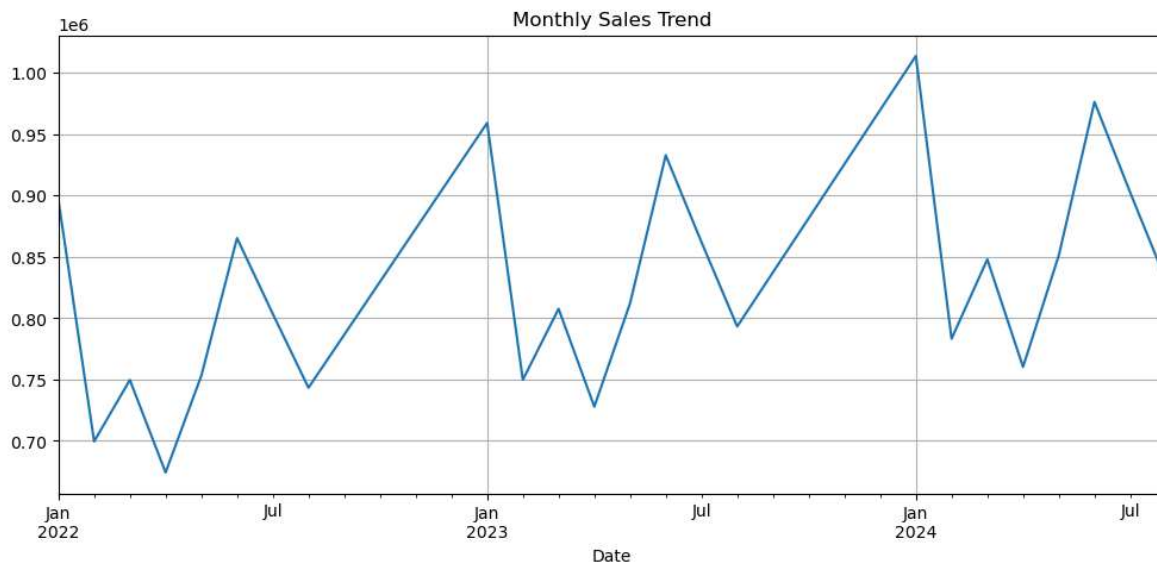
	Sales Person	Country	Product	Date	Amount	Boxes Shipped	Year	Month	MonthName	Quarter
--	--------------	---------	---------	------	--------	---------------	------	-------	-----------	---------



Following are few trends in the data

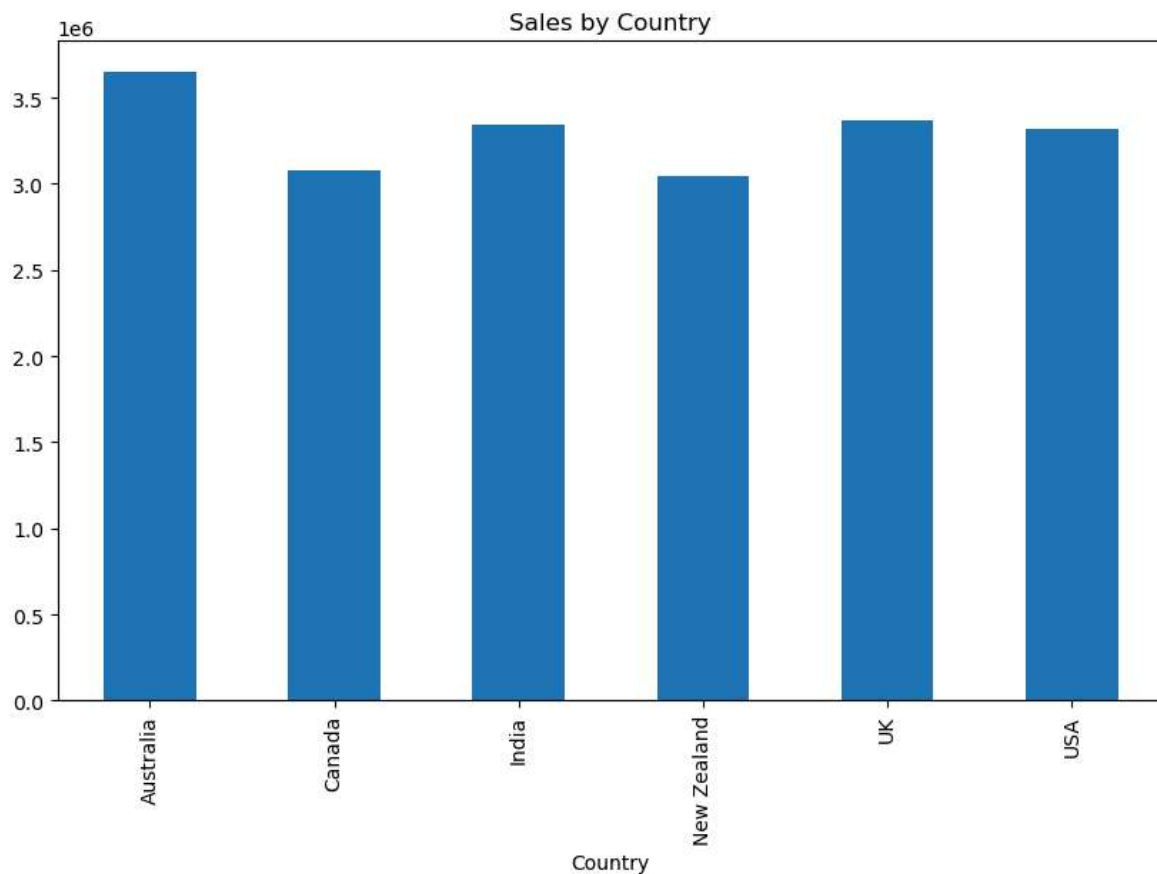
```
In [21]: # Following are few trends in the data
monthly_sales = df.groupby(df['Date'].dt.to_period('M'))['Amount'].sum()
monthly_sales.plot(kind='line', figsize=(12,5), title='Monthly Sales Trend', gri
```

```
Out[21]: <Axes: title={'center': 'Monthly Sales Trend'}, xlabel='Date'>
```



```
In [22]: sale_country= df.groupby('Country')['Amount'].sum()
sale_country.plot(kind='bar',figsize=(10,6), title='Sales by Country')
```

```
Out[22]: <Axes: title={'center': 'Sales by Country'}, xlabel='Country'>
```



```
In [23]: df.head()
```

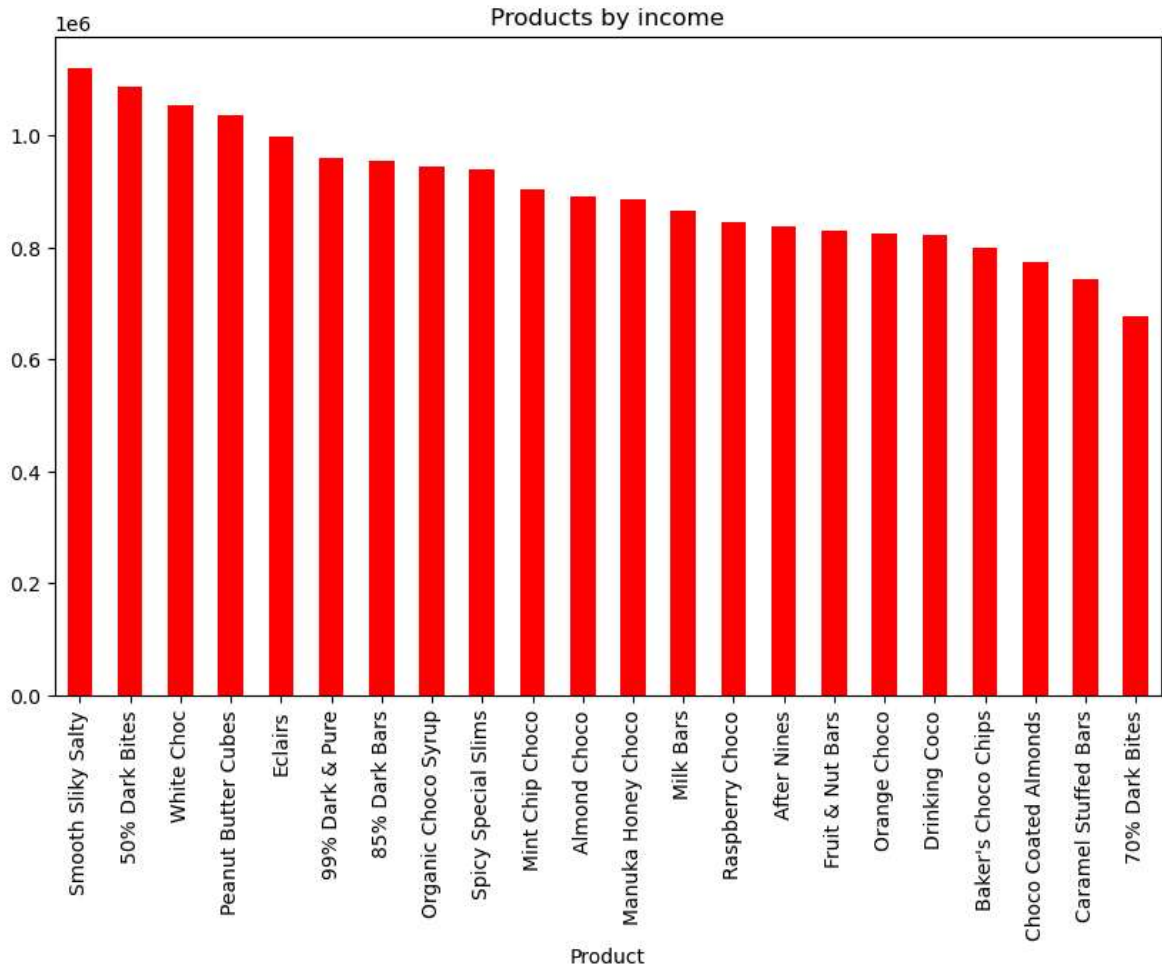
```
df['Product'].nunique()
```

Out[23]: 22

Most income generating products

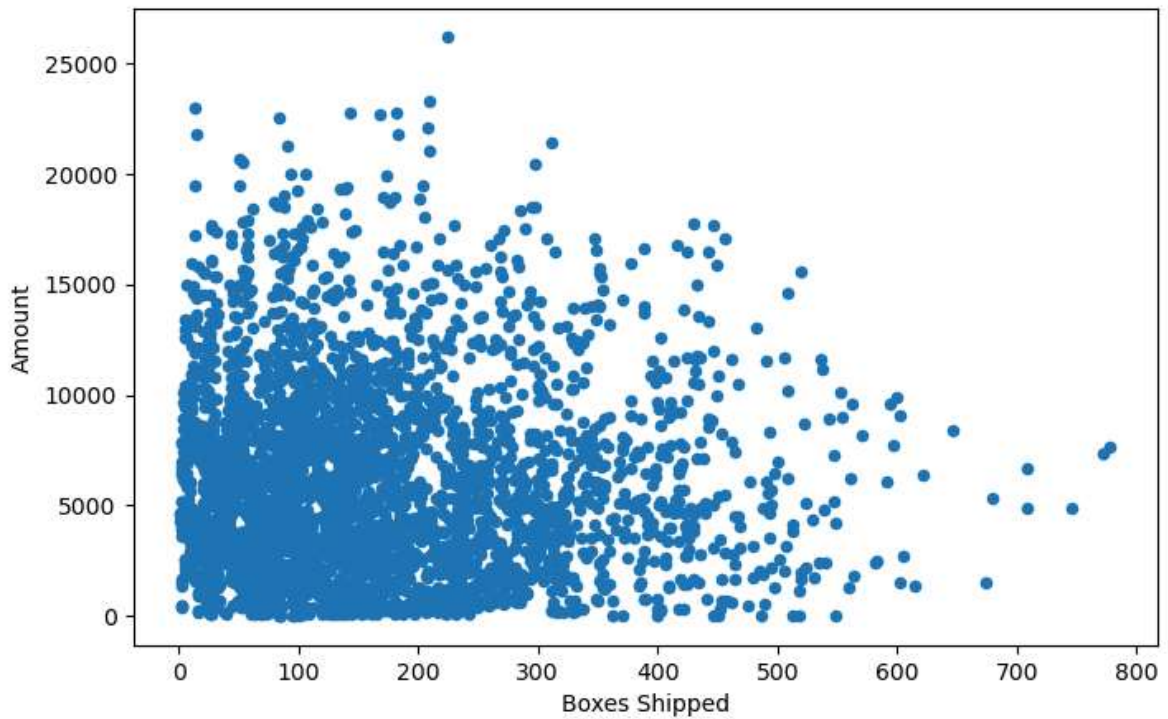
```
In [24]: products=df.groupby('Product')['Amount'].sum().sort_values(ascending=False)
products.plot(kind='bar', figsize=(10,6),color='red',title='Products by income')
```

Out[24]: <Axes: title={'center': 'Products by income'}, xlabel='Product'>



```
In [73]: df.plot(kind='scatter', x='Boxes Shipped', y='Amount', figsize=(8,5))
```

Out[73]: <Axes: xlabel='Boxes Shipped', ylabel='Amount'>



There is no linear relation btw amount and boxes shipped. Hence a regression model is not fit here.

```
In [53]: from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

X = df[['Amount', 'Boxes Shipped']]
```

```
In [39]: scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
In [41]: kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X_scaled)
df['Cluster'] = kmeans.labels_
```

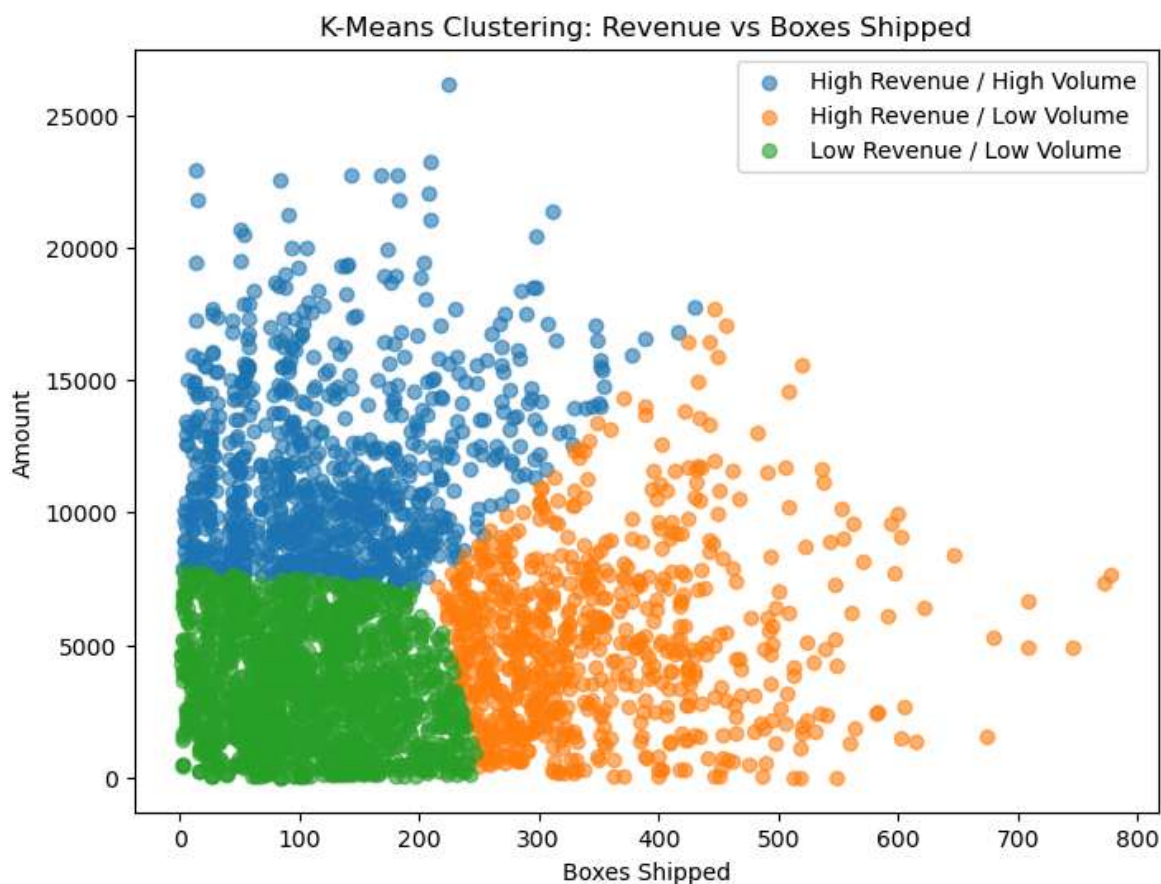
```
In [49]: df.groupby('Cluster')[['Amount', 'Boxes Shipped']].mean()
cluster_labels = {
    0: "Low Revenue / Low Volume",
    1: "High Revenue / High Volume",
    2: "High Revenue / Low Volume"
}
df['ClusterLabel'] = df['Cluster'].map(cluster_labels)
```

```
In [57]: import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

for label, group in df.groupby('ClusterLabel'):
    plt.scatter(
        group['Boxes Shipped'],
        group['Amount'],
        label=label,
        alpha=0.6
    )
```

```
plt.xlabel('Boxes Shipped')
plt.ylabel('Amount')
plt.title('K-Means Clustering: Revenue vs Boxes Shipped')
plt.legend()
plt.show()
```



In [61]: `df.head()`

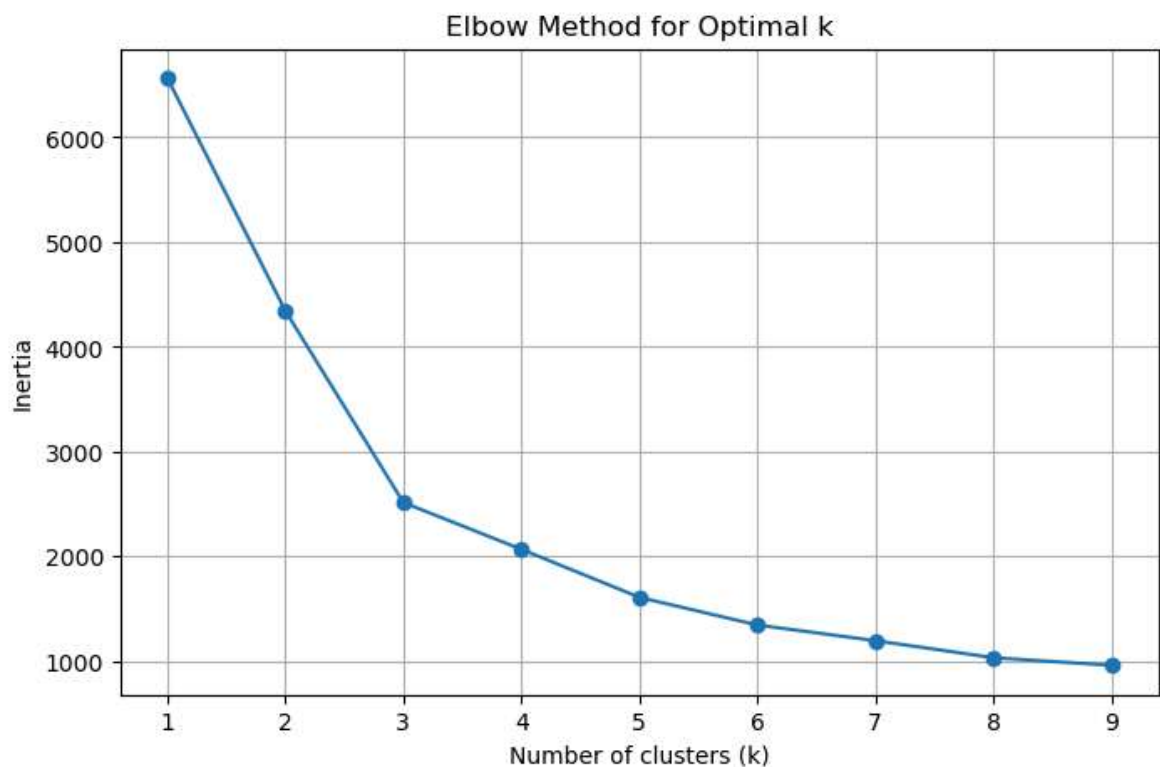
Out[61]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped	Year	Month	MonthName
1012	Dotty Strutley	USA	Raspberry Choco	2022-01-03	3437.0	46	2022	1	January
253	Marney O'Brien	Australia	Smooth Sliky Salty	2022-01-03	3745.0	170	2022	1	January
89	Andria Kimpton	Canada	Caramel Stuffed Bars	2022-01-03	7154.0	348	2022	1	January
187	Dotty Strutley	UK	99% Dark & Pure	2022-01-03	9310.0	282	2022	1	January
185	Camilla Castle	UK	White Choc	2022-01-03	9982.0	187	2022	1	January

```
In [67]: inertia = []
K = range(1, 10)

for k in K:
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X_scaled)
    inertia.append(km.inertia_)
```

```
In [71]: plt.figure(figsize=(8,5))
plt.plot(K, inertia, marker='o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method for Optimal k')
plt.xticks(K)
plt.grid(True)
plt.show()
```



The elbow method shows a clear bend at $k = 3$, indicating that three clusters provide the best balance between model simplicity and cluster compactness. Increasing k beyond 3 yields diminishing returns, so $k = 3$ was selected as the optimal number of clusters.

Next Steps

Build a time-series forecast to predict future sales trends

Develop an interactive dashboard using Plotly, Power BI, or Tableau